MILITARY SPECIFICATION SHEET

CABLE ASSEMBLIES, NONPRESSURE PROOF, FIBER OPTIC, TEST, METRIC

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-49292.

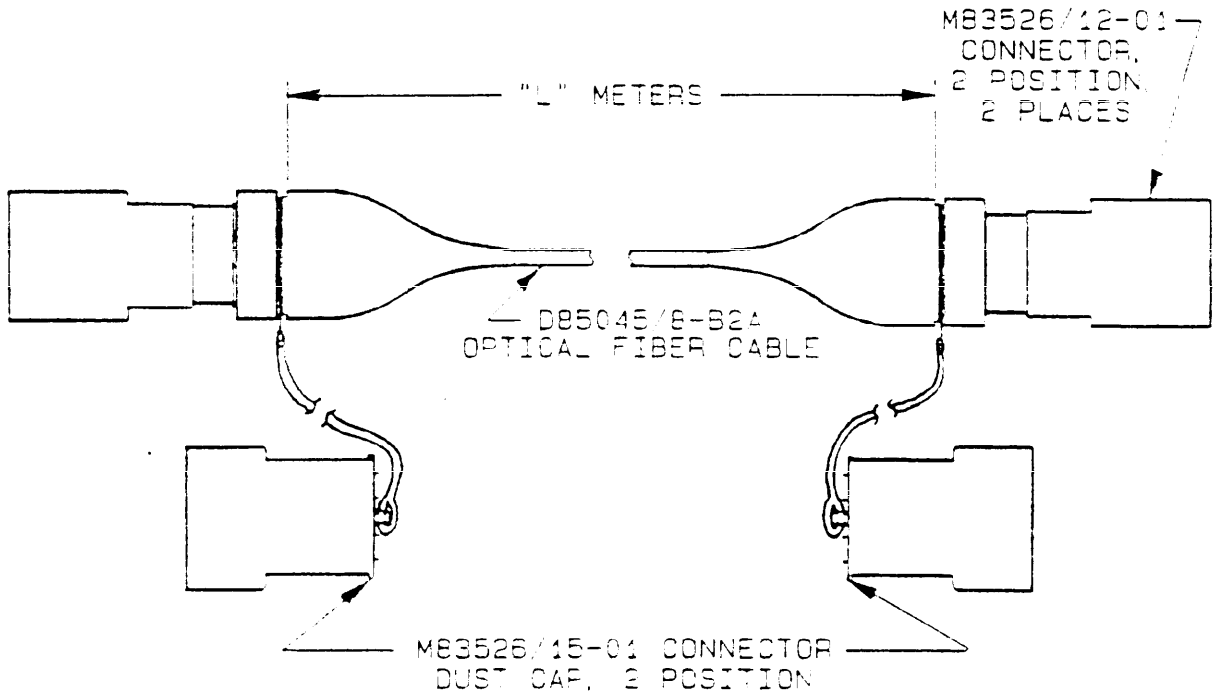


FIGURE 1. Outline drawing, cable assembly (hermaphroditic) fiber optic, test.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:01Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Color: Colors for coding of fibers within the cable assembly shall be blue and orange (see table I).

TABLE I. Color-coded termination configuration.

Termini	Buffer color	Termini
1	Blue	2
2	Orange	1

Component parts:

Connectors shall be in accordance with the following:

M83526/12-01, two required.

Dust cover assembly: M83526/15-01, two required.

Cable: D85045/8-B2A.

Assembly: In accordance with figure 1, and tables I and II.

Length: In accordance with table II.

Weight: In accordance with table II.

Performance:

Optical requirements:

Insertion loss (maximum assembly attenuation): 1/

Measurements made at 1290 nm ±10 nm and at 25°C. Test method B1.

Part or Identifying Number (PIN)	
M49292/7-01	2.75 dB ±0.5 dB
M49292/7-02	8.25 dB ±0.5 dB

Optical fiber bandwidth: Not applicable.

Crosstalk: Not applicable.

Cable seal flexing: Not applicable.

Thermal shock: Not applicable.

1/ Test cable insertion loss shall include cable loss, connector loss and the loss of any required optical attenuation element(s) within each connector.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:01Z
Check the source to verify that this is the current version before use.

Cable retention: Not applicable.

Fluid immersion: Not applicable.

Temperature range:

Operating: 0°C to +52°C.

Storage: -46°C to +71°C.

Dust: Not applicable.

Storage temperature: Not applicable.

Life test: Not applicable.

Salt fog: Not applicable.

PIN: PINs shall be in accordance with table II.

Temperature cycling: The test cable assembly shall be tested in accordance with the following:

Low temperature: MIL-STD-810, method 502, procedures I and II.

High temperature: MIL-STD-810, method 501, procedures I and II (constant temperature).

The minimum soak time at each storage and operating temperature shall be 2 hours.

TABLE II. PINs.

Military PIN M49292/7-	Length "L" (meters)	Weight (kg) maximum
01	10.0 ±0.5	3.2
02	10.0 ±0.5	3.2

First article inspection shall consist of performing the inspections and optical tests specified in table III herein, in the sequence shown.

Group A inspection shall consist of performing the inspections and optical tests specified in table IV herein, in the sequence shown.

Group B inspection: Not applicable.

Group C inspection shall consist of performing the inspections and optical tests specified in the table V herein, in the sequence shown.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:01Z
Check the source to verify that this is the current version before use.

TABLE III. First article inspection.

Inspection	Sample quantity
Group I	
Visual and mechanical	6 units
Insertion loss	6 units
Group II	
Not applicable	
Group III	
Temperature cycling	2 units
Humidity	1 unit
Vibration	1 unit
Low pressure (altitude)	1 unit

TABLE IV. Group A inspection.

Inspection
Visual and mechanical
Insertion loss

TABLE V. Group C inspection.

Inspection	Sample quantity
Subgroup I (3-month interval)	
Temperature cycling	2 units
Subgroup II (6-month interval)	
Humidity	1 unit
Vibration	1 unit
Subgroup III (12-month interval)	
Low pressure (altitude)	1 unit

Revision letters are not used to denote changes due to the extensiveness of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:01Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI, SC
Navy - AS, CG, EC, MC
Air Force - 99
DLA - ES

User activities:
Army - AV
Navy - CG, MC

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 6020-0002-7)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:01Z
Check the source to verify that this is the current version before use.